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Heterogeneous network for Hierarchical Fine-Grained Domain Fake News Detection

Yue Wang [a], Shizhong Yuan [a], Weimin Li [a,*], Yifan Feng [a], Xiao Yu [a], Fangfang Liu [a], Can Wang [b], Quanke Pan [c]

[a] School of Computer Engineering and Science, Shanghai University, Shanghai 200444, Shanghai, China
[b] School of Information and Communication Technology, Griffith University, Gold Coast 4222, Queensland, Australia
[c] School of Mechatronic Engineering and Automation, Shanghai University, Shanghai 200444, Shanghai, China

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ABSTRACT

Fake news on social media has significant negative consequences for individuals and society. However, existing multi-domain detection methods exhibit two primary limitations: dependence on precise domain annotation and inherent bias arising from single-domain categorization. To address these challenges, this paper introduces the Domain-Specific Narrow-Coverage Tree-Based Taxonomy (DNT^2), which enables more precise domain classification and domain relationship elucidation through refined categories. The constructed dataset is annotated with multiple labels by Large Language Models (LLMs), mitigating reliance on manual efforts and reducing annotation costs while maintaining annotation quality. Furthermore, a Hierarchical Fine-Grained Domain (HFGD) Fake News Detection Method is proposed, which explicitly employs a heterogeneous network to model multi-relationships. This method can mitigate domain bias and comprehensively capture news diversity and domain interactions. Specifically, domain cohesion based on news semantics is designed to reflect the relevance of news within a domain. News items are integrated as intersection nodes into the tree structure of multilevel domains to construct the heterogeneous network. Graph representation learning then fuses directly or indirectly connected news and domain information during feature enhancement. Finally, a composite loss is designed for news and domain node classification. HFGD captures potential differences and commonalities in domains and enhances label adaptation through domain interactions. Experiments on our dataset demonstrate that HFGD outperforms state-of-the-art methods by 1.08% and 0.91% in overall accuracy and macro-F1 score, respectively. Specifically, in the education and military domains with limited sample sizes, HFGD achieves 5.74% and 4.1% improvements in macro-F1 score over the second-best method. The results demonstrate our method's effectiveness in mitigating domain bias and enhancing detection performance, providing valuable insights for practical multi-domain fake news detection systems.

1. Introduction

Online social media platforms, such as Twitter and Weibo, have significantly enhanced how people access and exchange information and become popular news sources (Hu, 2023; Stokel-Walker, 2022). Despite their benefits, the openness and anonymity

* Corresponding author.
E-mail addresses: wyofficial@shu.edu.cn (Y. Wang), szyuan@shu.edu.cn (S. Yuan), wmli@shu.edu.cn (W. Li), fengyffyf@shu.edu.cn (Y. Feng), yuuuu@shu.edu.cn (X. Yu), ffliu@shu.edu.cn (F. Liu), can.wang@griffith.edu.au (C. Wang), panquanke@shu.edu.cn (Q. Pan).

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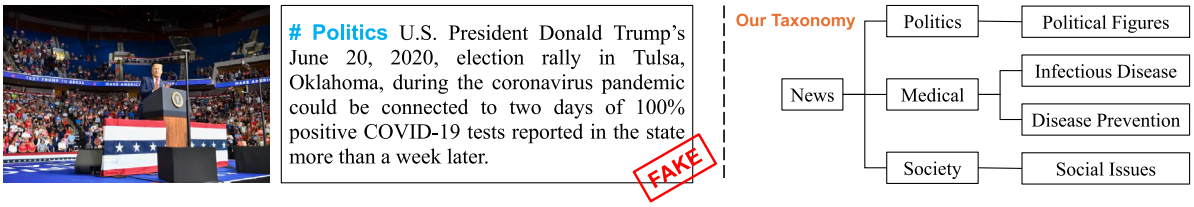


Fig. 1. An example of news encompassing multiple domains.

of these platforms also enable the rapid dissemination of maliciously crafted fake content (Taylor et al., 2023). For instance, during the 2016 U.S. presidential election and the COVID-19 pandemic, there was a proliferation of fake news and hate speech (Allcott & Gentzkow, 2017; Varshney & Vishwakarma, 2021) related to political propaganda, health-threatening, etc. These deceptive contents have become a pressing issue that poses serious threats to political security and social stability (Allcott & Gentzkow, 2017; González-Bailón et al., 2023; Varshney & Vishwakarma, 2021).

These news encompass various domains such as politics, healthcare, and social life. While several studies have proposed methods for multi-domain fake news detection, these approaches have several challenges:

(1) ***Domain label annotation is costly, and models lack robustness against noise.*** Present datasets require the manual annotation of domain labels for a substantial number of samples (Nan et al., 2021). This process is not only time-consuming but also demands significant human resources. It presents a problem that remains unresolved under existing conditions. Although automatic tools, such as Large Language Models (LLMs), can partially alleviate the burden, their accuracy is typically lower than manual efforts. This results in suboptimal performance on noisy datasets.

(2) ***Domain labels are sparse and biased, and models lack explicit multi-relationship modeling.*** Present research often categorizes each news item into a single domain without considering instances where samples pertain to multiple domains simultaneously (Mosallanezhad et al., 2022; Nan et al., 2021; Yue et al., 2023; Zhu et al., 2023). Although Zhu et al. (2023) advocate for supplementing domain labels, the overall labels remain sparse and fail to capture the diversity of real-world news adequately. Furthermore, it is intuitive that news items within the same domain should exhibit intrinsic connections, and domains should also be interrelated through common news items. However, these approaches fail to consider inter-domain relationships and lack multi-relational modeling.

As is shown in Fig. 1, this news item concerns a fake report on the surge of COVID-19 cases in Oklahoma. This example spans multiple domains, including *Politics*, *Medical*, and *Society*. Additionally, the number of infections and rally leading to a 100% positive rate implicates subdomains such as *Infectious Disease* and *Disease Prevention* within the *Medical*. This illustrates that a single news item may be simultaneously involved in several domains and multiple subdomains within a given one. Consequently, it is imperative to advance research on multi-domain fake news detection that reflects real-world situations more accurately (Sheng, Cao, Bernard, et al., 2022).

To address the above challenges, we first introduce a **D**omain-Specific **N**arrow-Coverage **T**ree-Based **T**axonomy (DNT^2) with the graphical representation of hierarchical relationships for categorizing news domains. This system clarifies the logical relationships between different domains through a hierarchical containment structure. It can minimize domain bias and be seamlessly transformed into a graph structure. Next, a multi-domain multi-label enhanced dataset is constructed based on DNT^2. It aggregates verified news from PolitiFact[1] and Snopes[2]. Each sample is annotated by LLM with multilevel domain labels, which adequately captures the diversity of real-world news and reduces the reliance on manual efforts. We further propose a **H**ierarchical **F**ine-**G**rained **D**omain (HFGD) Fake News Detection Method, that explicitly models multi-relationships through graph structure to capture potential cross-domain interaction patterns of news. Specifically, HFGD constructs a heterogeneous network based on the tree structure of DNT^2. This network employs the distribution of Shannon diversity index to measure the semantic cohesion of news within each domain. News are introduced as intersection nodes, and multi-relationships are constructed by similarity and domain labels. Subsequently, graph representation learning is performed on this heterogeneous network to fuse direct or indirect news and domain features for the nodes. On the one hand, the enriched node information can be better categorized. With the assistance of news in the relevant domain, on the other hand, noise caused by automatic annotation can be reduced to a certain extent, so as to improve the robustness of the model. Finally, a composite loss containing domain and news node classification is designed to ensure that distinctions between different domains are sufficiently significant. The main contributions are summarized as follows:

1. We introduce the Domain-Specific Narrow-Coverage Tree-Based Taxonomy (DNT^2) to mitigate the sparsity of domain labels and clarify the relationships between domains. It allows the model to learn more robust and generalized news representations, improving its ability in detection.
2. We propose the Hierarchical Fine-Grained Domain (HFGD) Fake News Detection Method, which expresses realistic news-domain multi-relationships through a heterogeneous network to reduce the domain bias and improve the adaptation to labels. The designed domain cohesion in HFGD can measure the structural properties of similar news in the same domain.

[1] https://www.politifact.com/
[2] https://www.snopes.com/

3. We construct a new multi-domain multi-label enhanced dataset. The results of the experiments demonstrate the effectiveness of the proposed method.

2. Related work

Fake news is defined as news whose content can be falsified (Shu et al., 2017; Vosoughi et al., 2018). This paper follows the same definition as in previous works, referring to stories or statements of an assertive nature as 'news'. Fake news detection is typically regarded as a binary classification task, with the objective of automatically classifying news content as true or false. To accomplish this, researchers have proposed a variety of detection techniques. In the following sections, these methods will be individually outlined.

2.1. Non-graph methods

Non-graph methods detect fake news by analyzing the content features of the news or by incorporating auxiliary information. Content-based methods mainly rely on news content features for classification (Shu et al., 2017). Since news content may contain visual information and text, many methods fuse information from different modalities. MVAE (Khattar et al., 2019) helps detection by learning shared representations of text and vision. MCAN (Wu et al., 2021) fuses visual and text information by designing a co-attention layer and learns more complex interactions between text and visual modalities. HMCAN (Qian et al., 2021) jointly models multimodal contextual information and hierarchical semantics of text. While other methods consider the consistency between different modalities. EM-FEND (Qi et al., 2021) explicitly models the inconsistency between visual entities and text entities. BMR (Ying et al., 2023) introduces a coarse judgment mechanism to explicitly measure contributions of different modalities, and reduces the dependence of detection results on this property by stripping cross-modal consistency learning. MSACA (Wang et al., 2024) improves text–image semantic consistency in the latent space through constructing hierarchical multi-scale images and adopts an attention module to select the critical embedding vectors. CAF-ODNN (Luvembe et al., 2024) uses bidirectional complementary attention to capture fine-grained cross-modal correlation modeling and incorporates specialized alignment and normalization components to optimize the fused representation.

Methods that incorporate auxiliary information fuse user characteristics (Dong et al., 2018; Zhang et al., 2021), propagation sequences (Liu & Wu, 2018), knowledge graphs (Dun et al., 2021), news environments (Sheng, Cao, Zhang, et al., 2022), etc. Dong et al. (2018) use the attention mechanism to fuse text and user information to improve the ability of fake news detection. BERT-EMO (Zhang et al., 2021) considers the relationship between dual emotions (publisher and social emotions) and reveals unique emotional signals. Liu and Wu (2018) model the news propagation process in social networks as a time series and capture the global and local information on it. KAN (Dun et al., 2021) uses multi-head attention to fuse text information, entity information in the knowledge graph, and entity context information. NEP (Sheng, Cao, Zhang, et al., 2022) observes the relationship between a given news and its news environment, providing a new perspective.

Recent studies have begun to explore using LLMs to provide multi-perspective information support for smaller models in fake news detection. For instance, CICAN (Yang et al., 2023) combines semantic feature learning through swarm intelligence with semantic structure mining based on knowledge, simultaneously utilizing ChatGPT to enhance knowledge extraction and entity linking. Hu et al. (2024) selectively draw insights from multi-perspective rationales generated by LLMs to enhance the performance of smaller language models in fake news detection. Yan et al. (2024) propose a knowledge-driven prompting strategy that enhances LLMs' semantic understanding and adaptability by building exploratory prompts and dynamically injecting external knowledge to address the challenge of insufficient training data. These approaches aim to enhance smaller models' capabilities by utilizing LLMs' broad knowledge and reasoning abilities.

2.2. Graph-based methods

Graph-based methods build graph structures (homomorphic or heterogeneous) and use Graph Neural Networks (GNN) for fake news detection. Generally, most work constructs graphs by user relationships and retweets on social media platforms. For example, Jiang et al. (2019) use the user associations in the friendship network to reconstruct the news-user network, thereby enhancing the news representation. GCAN (Lu & Li, 2020) uses user features and propagation sequences to construct a graph to simulate the potential interactions between users, and uses Graph Convolutional Networks (GCN) to learn the graph-aware representation of user interactions. Bi-GCN (Bian et al., 2020) models the news propagation process as a graph structure, and learns the propagation and diffusion patterns of news through a bidirectional propagation mechanism. Shu et al. (2019) consider the publishing relationship between publishers and news, the propagation relationship between users and news, and the following relationship between users to construct a heterogeneous information network, then use matrix decomposition to obtain the embedded representation of each news node for fake news detection. GAMC (Yin et al., 2024) combines graph autoencoders with masking and contrastive learning for unsupervised fake news detection, utilizing news content and propagation context as self-supervised signals.

There are also some works that build graphs by integrating information from knowledge graphs with news content. For example, KMGCN (Wang et al., 2020) builds a heterogeneous information network containing text, image, and entity information in knowledge graphs, and uses GCN to fuse these information. Hu et al. (2021) compare the entity information extracted from news content with that in the knowledge graph based on the constructed directed heterogeneous document graph to capture the consistency between

news content and the knowledge base. AKA-Fake (Zhang, Zhang, et al., 2024) employs reinforcement learning to adaptively generate knowledge subgraphs that are closely related to news content, and integrates them via heterogeneous graph learning.

Recent studies have begun to explore more sophisticated graph construction approaches for capturing different aspects of fake news detection. FCN-LP (Zhao et al., 2023) uses CLIP to extract cross-modal features, constructs the latent connections between tweets, and models the positive and negative correlations between tweets through feature contextualization and label propagation networks. HeteroSGT (Zhang, Ma, et al., 2024) constructs a heterogeneous graph involving news articles, entities, and topics, then combines the random walk and restart (RWR) algorithm to extract subgraphs centered on each news for detection. MSynFD (Xiao et al., 2024) captures the complex syntactic relations in fake news by combining the syntactic dependency graph and the multi-hop subgraph aggregation mechanism, and designs a relative position-aware Transformer model and a keyword debiasing module to improve the accuracy and robustness of fake news detection.

2.3. Multi-domain methods

Multi-domain methods improve the performance by jointly modeling news in multiple domains. Some studies focus on the acquisition of domain-invariant shared features. For example, EANN (Wang et al., 2018) removes event-specific features by generating adversarial networks. REAL-FND (Mosallanezhad et al., 2022) integrates features such as news content, user comments, and user-news interactions, then uses reinforcement learning to obtain domain-independent features. UCD-RD (Ran & Jia, 2023) constructs instance-level and prototype-level contrastive learning to align features and uses a cross-attention mechanism to learn domain-invariant features. EMT (Bazmi et al., 2024) learns domain-invariant and domain-specific representations in the news and integrates entity contextual information from external knowledge sources to improve the generalization ability.

While some other studies concentrate on integrating domain-specific features. For example, MDFEND (Nan et al., 2021) uses domain gates to aggregate feature representations obtained from different expert networks and uses the aggregated features for fake news detection. M^3FEND (Zhu et al., 2023) uses domain adaptors to adaptively aggregate cross-view representations. CANMD (Yue et al., 2022) uses label distribution offset correction and conditional distribution contrast adaptation between the source and target domains for early detection. MetaAdapt (Yue et al., 2023) guides knowledge transfer from the source domain to the target domain through meta-learning.

Recent research has explored methods that do not rely on large labeled datasets. For example, Silva et al. (2024) fuses multiple weak signals to encode news knowledge, utilizes self-supervised learning with a teacher-student architecture for detection, and employs the backward dataset construction strategy to alleviate selection bias.

Although numerous works have investigated the application of GNN in fake news detection and recent studies have begun to explore more sophisticated graph construction approaches, existing methods still primarily focus on single-domain scenarios with limited relationship types (e.g., retweet networks, knowledge graphs, or syntactic dependencies). There is a notable lack of comprehensive frameworks that model diverse relationships in multi-domain fake news detection. To address this gap, our method incorporates multi-relationship modeling based on a heterogeneous network to improve cross-domain detection performance.

3. Motivation

The proliferation of fake news across various domains poses a significant threat to societal trust and informed decision-making processes. However, existing methods for multi-domain fake news detection depend heavily on precise manual domain annotations and exhibit poor performance when noise is introduced into data labels. Moreover, most of these methods are limited to a single label or relationship. These oversimplified patterns fail to capture the hierarchical dependencies and interactions of news in real-world scenarios. These issues limit the practical applicability of existing approaches in multi-domain settings, highlighting the urgent need for more advanced detection methodology to address these challenges.

This paper proposes a novel method for multi-domain fake news detection, with the following objectives:

- To introduce a hierarchical domain taxonomy that provides fine-grained categorization of news content. This taxonomy aims to reduce domain bias and enhancing diversity through detailed classification while supporting rich domain interactions. The fine-grained structure also facilitates more accurate domain-specific feature learning and effective knowledge transfer across related domains.
- To develop a new robust detection framework that mitigates the impact of noisy domain labels and capturing diverse interactive relationships. By constructing a heterogeneous graph that incorporates both content-based relationships and domain-level dependencies, the framework can capture interactions across domains while being resilient to label inaccuracies. It employs graph-based message passing to transfer knowledge between related domains, thereby improving detection performance in domains with noisy training data.
- To leverage domain knowledge and shared patterns through carefully designed graph structures. Each domain can benefit from relevant knowledge in related domains while maintaining its specific characteristics by capturing and utilizing domain relationships and shared characteristics in the graph structure. This approach focuses on transferring learned representations and patterns that help characterize fake news, enabling more effective cross-domain knowledge sharing and enhancing the detection capability in individual domains.

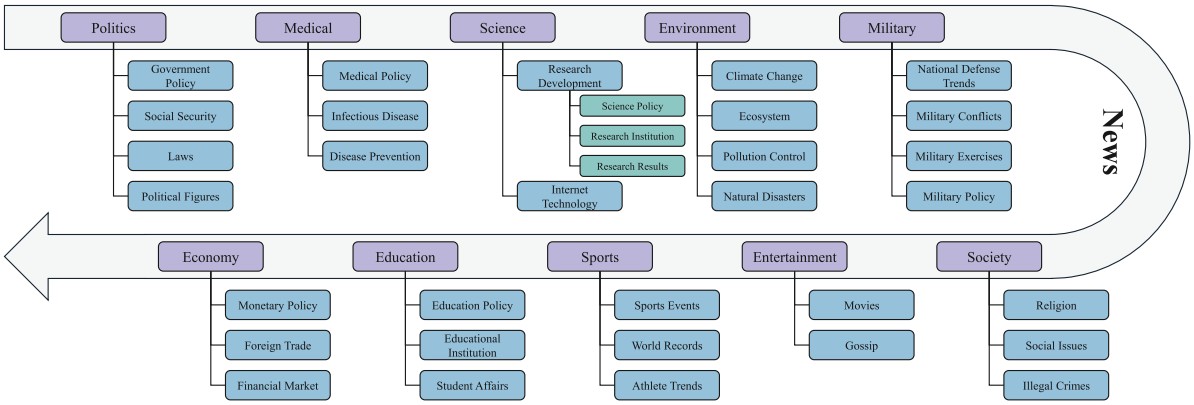


Fig. 2. The tree structure of DNT^2.

4. Preliminaries

4.1. Multi-domain news heterogeneous graph

News articles naturally span multiple domains and exhibit complex relationships both within and across domains. To capture these rich semantic connections and domain knowledge, we construct a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ for multi-domain fake news detection. The node set $\mathcal{V} = \{v_1, v_2, \ldots, v_{|\mathcal{V}|}\}$ consists of two types of nodes mapped by function $f(\mathcal{V}) \rightarrow \boldsymbol{\Phi}$, where $\boldsymbol{\Phi} = \{\text{News}, \text{Domain}\}$:

- Domain nodes: Domains from the complete domain set Ω are from a hierarchical taxonomy. Each domain node is represented by a subgraph structure $\mathcal{G}_d = (\mathcal{V}_d, \mathcal{E}_d)$ constructed from domain-specific news articles to capture their internal relationships.
- News nodes: Each news item from dataset $\mathcal{S} = \{\mathcal{P}_1, \mathcal{P}_2, \ldots, \mathcal{P}_{|\mathcal{S}|}\}$ is represented as $\mathcal{P}_i = (\mathcal{C}_i, \mathcal{Y}_i^{dom}, y_i)$, where $\mathcal{C}_i = \left\{t_1^i, t_2^i, \ldots, t_{|\mathcal{C}_i|}^i\right\}$ represents content tokens, $\mathcal{Y}_i^{dom} \subseteq \Omega$ represents domain labels and $y_i \in \{0, 1\}$ indicates whether news is real or fake.

The edge set $\mathcal{E} = \left\{ e_{ij} = (v_i, v_j) \middle| v_i, v_j \in \mathcal{V} \right\}$ contains three types of bidirectional relationships mapped by function $g(\mathcal{E}) \rightarrow \boldsymbol{\Psi}$, where $\boldsymbol{\Psi} = \{\text{News-News}, \text{News-Domain}, \text{Domain-Domain}\}$:

- News-News edges: Connect related news items based on content similarity.
- News-Domain edges: Link news to their associated domains from $\mathcal{Y}_i^{dom}$.
- Domain-Domain edges: Capture hierarchical relationships between domains in the taxonomy.

4.2. Multi-domain fake news detection

Given the constructed heterogeneous graph $\mathcal{G}$, we formulate multi-domain fake news detection as a node classification problem. For each news node $\mathcal{V}_{news} \subset \mathcal{G}$ extracted from $\mathcal{S}_{train} = \left\{(\mathcal{C}_i, \mathcal{Y}_i^{dom}, y_i)\right\}_{i=1}^{|\mathcal{S}_{train}|}$, our goal is to learn a function $f : \mathcal{V} \times \mathcal{E} \rightarrow \hat{y}$ that can effectively predict the veracity of news. The function f integrates the content information from news texts, structural relationships in the heterogeneous graph, and cross-domain knowledge from related domains. This multi-faceted approach facilitates robust fake news detection by leveraging both local content features and global domain dependencies captured in the graph structure.

5. Methodology

5.1. Architecture

In order to reduce domain bias and reflect the logical relationship between domains, this paper proposes DNT^2 based on the existing domain categories and HFGD fake news detection method. The constructed tree-based domain classification system is depicted in Fig. 2. The architecture of the proposed HFGD based on DNT^2 is shown in Fig. 3, which is meticulously designed to make use of the news-domain multi-relationships through explicit modeling. The news data sequentially undergoes (1) network construction, (2) feature enhancement, and (3) prediction with auxiliary domain classification to obtain the final result.

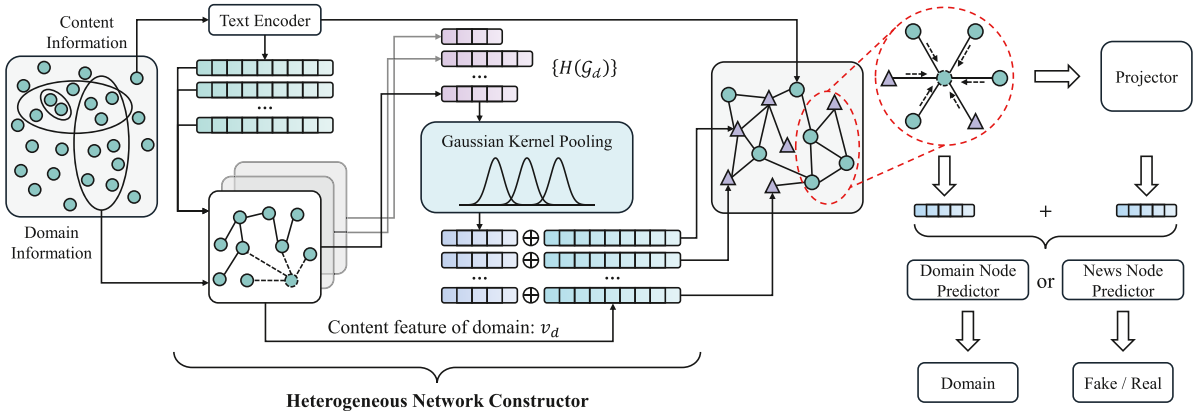


Fig. 3. Overview of the Hierarchical Fine-Grained Domain (HFGD) Fake News Detection Method. The raw data is sequentially subjected to network construction, feature enhancement, and auxiliary classification to obtain the final prediction results.

5.2. Tree-based domain taxonomy

The proposed taxonomy structure is constructed based on general domain knowledge and standard categorization practices. Specifically, we first identify 10 principal domain labels by synthesizing widely-accepted content categories from existing literature (Nan et al., 2021; Vosoughi et al., 2018): ***Politics***, ***Economy***, ***Medical***, ***Education***, ***Science***, ***Sports***, ***Environment***, ***Military***, ***Entertainment***, ***Society***. To establish a more fine-grained classification system, we further develop the hierarchical structure by following the principles of Linnaean taxonomy (Linnaeus, 1758) and referring to issue categorization schemes used by fact-checking organizations like PolitiFact. This domain-agnostic approach results in a comprehensive classification system with 44 distinct domain categories across multiple levels, which can be generally applied to analyze fake news across different contexts and datasets.

Based on the tree classification of domain labels in DNT[2], let there be ℓ levels. The set of major domains is defined as $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$, where d_i denotes the ith major domain. Each domain d_i has a direct subset $\mathcal{D}_i = \left\{d_{i1}, d_{i2}, \dots, d_{in_i}\right\}$, where d_{ij} denotes the jth subdomain of d_i, and n_i denotes the number of subdomains of d_i. Similarly, each subdomain d_{ij} also has a direct subset, denoted as $\mathcal{D}_{ij} = \left\{d_{ij1}, d_{ij2}, \dots, d_{ijn_{ij}}\right\}$. This process is repeated until the final level is reached. Therefore, the set of all domain labels Ω can be expressed as:

$$\Omega = \mathcal{D} + \underbrace{\sum_i \left(\mathcal{D}_i + \sum_j \left(\mathcal{D}_{ij} + \cdots + \sum_k \mathcal{D}_{ij\dots k}\right)\right)}_{\ell-1}, \tag{1}$$

where the addition and summation operations are applied to sets to determine their concatenation.

5.3. Network construction

5.3.1. Nodes features

News Nodes Feature. The purpose here is to extract the semantic representation of news content as the features of news nodes. The sentence embeddings extracted by BERT are of a poor quality and not suitable for the task of semantic similarity computation according to Reimers and Gurevych (2019), so their proposed Sentence-BERT is employed to encode the tokens $\left\{\boldsymbol{t}_1, \boldsymbol{t}_2, \dots, \boldsymbol{t}_{|\mathcal{C}|}\right\}$ of the news content:

$$\boldsymbol{r}_i^{news} = \text{SBERT}\left(\left\{\boldsymbol{t}_1, \boldsymbol{t}_2, \dots, \boldsymbol{t}_{|\mathcal{C}|}\right\}\right), \tag{2}$$

where $\boldsymbol{r}_i^{news} \in \mathbb{R}^O$, O represents the feature dimension of news nodes. The features $\left\{\boldsymbol{r}_i^{news}\right\}_{i=1}^{|\mathcal{S}|}$ of all news are concatenated to $\boldsymbol{X}^{news} \in \mathbb{R}^{|\mathcal{S}| \times O}$.

Domain Nodes Feature. The purpose here is to use all news representations in the same domain to obtain the representations of domain itself. For a specific domain d, the set of samples labeled with it is obtained:

$$\Gamma_d = \left\{ \left(\mathcal{C}_i = \left\{\boldsymbol{t}_1^i, \boldsymbol{t}_2^i, \dots, \boldsymbol{t}_{|\mathcal{C}_i|}^i\right\}, \mathcal{Y}_i^{dom}, y_i\right) \middle| d \in \mathcal{Y}_i^{dom} \right\}. \tag{3}$$

Eq. (2) is utilized to obtain the set of semantic representations of news associated with the domain d:

$$R_d = \left\{ \boldsymbol{r}_1^{news}, \boldsymbol{r}_2^{news}, \dots \boldsymbol{r}_{|R_d|}^{news} \middle| \Gamma_d \right\}. \tag{4}$$

To derive the structural feature of news within the same domain, news items associated with the domain d are organized into a bidirected weighted homogeneous network $\mathcal{G}_d = (\mathcal{V}_d, \mathcal{E}_d)$. Within the same domain, news articles naturally form interconnected clusters as they discuss related events or issues. The network construction captures these intrinsic relationships, where news covering the same subject matter often demonstrates similar patterns in terms of content. Specifically, each sample in R_d corresponds to a node in $\mathcal{G}_d$, and weights $w_{e_{ij}^d}$ are assigned to each edge $e_{ij}^d \in \mathcal{E}_d$ by computing the cosine similarity of the features of the two connected nodes v_i^d, v_j^d:

$$w_{e_{ij}^d} = \mathrm{sim}\left(v_i^d, v_j^d\right). \tag{5}$$

This similarity helps capture the inherent relationships between news articles within the same domain.

When the proportion of removed components exceeds a threshold, the network fragments into smaller, disconnected parts. This isolation reflects the correlation among news items within this domain and allows insights into how different news items influence the overall domain, i.e., domain cohesion. Specifically, first compute the weighted betweenness centrality (Brandes, 2008) of each node in $\mathcal{G}_d$:

$$c_B(v) = \sum_{v_s, v_t \in \mathcal{V}} \frac{\delta\left(v_s, v_t \middle| v\right)}{\delta\left(v_s, v_t\right)}, \tag{6}$$

where $\delta\left(v_s, v_t\right)$ denotes the number of shortest paths from v_s to v_t that considering the edge weights, $\delta\left(v_s, v_t \middle| v\right)$ denotes the number of these shortest paths that pass through node v, and $v \notin \left\{v_s, v_t\right\}$. Betweenness centrality (Brandes, 2008) represents the degree of interaction between nodes and is related to the connectivity of the network. Removing nodes with high betweenness centrality is more likely to disconnect the graph (Holme et al., 2002). Therefore, the news nodes in $\mathcal{G}_d$ are removed sequentially based on the order of weighted betweenness centrality from highest to lowest. A modified Shannon diversity index (Zitnik et al., 2019) is employed to measure the fragmentation in the remaining network components. The connectivity component is utilized as a network component:

$$H_i\left(\mathcal{G}_d\right) = -\frac{1}{\log\left|\mathcal{V}_d\right|} \sum_j p_j \log p_j, \tag{7}$$

where $p_j = N_{comp_j} / \left|\mathcal{V}_d\right|$ denotes the proportion of nodes in $\mathcal{G}_d$ belonging to the jth network component, with $H_i\left(\mathcal{G}_d\right)$ taking a value between 0 and 1. By sequentially removing nodes, a list of Shannon diversity index $\left\{H_i\left(\mathcal{G}_d\right)\right\}_{i=1}^{|\mathcal{V}_d|}$ for $\mathcal{G}_d$ can be obtained.

However, since the number of news items varies across domains, it is not always suitable for networks with fixed input dimensions. Therefore, a Gaussian kernel pooling method (Sheng, Cao, Zhang, et al., 2022; Xiong et al., 2017) is employed to standardize the length of the lists of Shannon diversity index across different domains into fixed-dimensional representations. The key idea is to project the varying-length lists onto a fixed number of predefined Gaussian kernels, where each kernel functions as a 'soft bin' centered at different positions with different widths. These kernels can capture the distribution of values at different scales. By aggregating the outputs of all values within each kernel, we derive a fixed-dimensional representation that preserves the statistical properties of the original value distribution. Specifically, assuming K kernels $\left\{\mathbf{N}_k\right\}_{k=1}^{K}$ are used, the output of the ith value in the list for the kth kernel is:

$$\mathbf{N}_k^i = \exp\left(-\frac{\left(H_i\left(\mathcal{G}_d\right) - \mu_k\right)^2}{2\sigma_k^2}\right), \tag{8}$$

where μ_k and σ_k denote the mean and standard deviation of the kth kernel, respectively. The output of the kth kernel is the sum of the outputs for each Shannon diversity index within this kernel:

$$\mathbf{N}_k = \sum_{i=1}^{|\mathcal{V}_d|} \mathbf{N}_k^i. \tag{9}$$

The structural feature of domain d is obtained by concatenating the outputs of all kernels and normalizing the result:

$$\boldsymbol{s}_d = \mathrm{Norm}\left(\bigoplus_{k=1}^{K} \mathbf{N}_k\right), \tag{10}$$

where $\boldsymbol{s}_d \in \mathbb{R}^K$, $\bigoplus$ and $\mathrm{Norm}(\cdot)$ represents concatenation and L2 normalization operation, respectively.

Additionally, to enrich the final domain features, the semantic information of news within the domain is also incorporated. Specifically, domain structural feature $\boldsymbol{s}_d$ and domain semantic feature $\boldsymbol{v}_d$ (obtained by averaging the semantic representations of news) are concatenated and fused using a linear layer:

$$\boldsymbol{r}_d^{dom} = \boldsymbol{W}\left(\boldsymbol{v}_d \oplus \boldsymbol{s}_d\right) + \boldsymbol{b}, \tag{11}$$

where $\boldsymbol{r}_d^{dom} \in \mathbb{R}^Q$, Q represents the feature dimension of domain nodes. For better understanding of this process, we present the pseudo-code in Algorithm 1. The features $\left\{\boldsymbol{r}_d^{dom}\right\}_{d=1}^{|\Omega|}$ of all domains are concatenated to $\boldsymbol{X}^{dom} \in \mathbb{R}^{|\Omega| \times Q}$.

Algorithm 1 Domain Feature Extraction

Input: A specific domain d, the set of samples Γ_d labeled with d
Output: Domain feature $\boldsymbol{r}_d^{dom}$ of domain d

Step 1: Obtain domain network $\mathcal{G}_d$
$\mathcal{V}_d \leftarrow \text{SBERT}\left(\Gamma_d\right)$
for each pair of news $\left(v_i^d, v_j^d\right)$ in $\mathcal{G}_d$ **do**
  $w_{e_{ij}^d} \leftarrow \text{sim}\left(v_i^d, v_j^d\right)$ according to Eq. (5)
  Set weight $w_{e_{ij}^d}$ of edge e_{ij}^d in $\mathcal{E}_d$
end for
$\mathcal{G}_d \leftarrow \left(\mathcal{V}_d, \mathcal{E}_d\right)$
Step 2: Obtain structural features
Calculate weighted betweenness centrality $c_B(v)$ according to Eq. (6) for all $v \in \mathcal{V}_d$
Sort nodes by $c_B(v)$ in descending order
for $i = 1$ to $\left|\mathcal{V}_d\right|$ **do**
  Remove node with highest $c_B(v)$ from $\mathcal{G}_d$
  Calculate connected components $\{comp_i\}$ for remaining network
  Sort connected components $\{comp_i\}$ by size in descending order
  $H\left(\mathcal{G}_d\right) \leftarrow \{\}$
  for each connected component $comp_i$ in $\mathcal{G}_d$ **do**
    Calculate $H_i\left(\mathcal{G}_d\right)$ according to Eq. (7)
  end for
  Append $H_i\left(\mathcal{G}_d\right)$ to $H\left(\mathcal{G}_d\right)$
end for
Step 3: Gaussian kernel pooling
for $k = 1$ to K **do**
  for $i = 1$ to $\left|H_i\left(\mathcal{G}_d\right)\right|$ **do** /* $\left|H_i\left(\mathcal{G}_d\right)\right| = \left|\mathcal{V}_d\right|$ */
    Calculate the output $\mathbf{N}_k^i$ of the ith H on the kth Gaussian kernel according to Eq. (8)
  end for
  $\mathbf{N}_k \leftarrow \sum_i \mathbf{N}_k^i$
end for
$\boldsymbol{s}_d \leftarrow \text{Norm}(\bigoplus_{k=1}^{K} \mathbf{N}_k)$
Step 4: Combine semantic and structural features
$\boldsymbol{v}_d \leftarrow \text{Mean}(\mathcal{V}_d)$
$\boldsymbol{r}_d^{dom} \leftarrow \boldsymbol{W}(\boldsymbol{v}_d \oplus \boldsymbol{s}_d) + \boldsymbol{b}$
return $\boldsymbol{r}_d^{dom}$

5.3.2. Edges

- **News-News Edge** aims to build a direct link between different news. For $\forall v_i^{news}, v_j^{news} \in \mathcal{G}$, calculate the cosine similarity s_{ij} of their features ($\boldsymbol{r}_i^{news}$, $\boldsymbol{r}_j^{news}$), the existence of the edge e_{ij} can be defined as:

$$e_{ij} = \begin{cases} 1, & \text{if } s_{ij} \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

  where τ is a predefined similarity threshold.
- **News-Domain Edge** aims to establish a direct connection between each news item and the domain to which it belongs. For any news item $\mathcal{P}_i$, with its corresponding news node v_i and the set of domain nodes $\left\{ v_d^{dom} \middle| d \in \mathcal{Y}_i^{dom} \right\}$, the existence of the edge e_{id} can be defined as:

$$e_{id} = \begin{cases} 1, & \text{if } d \in \mathcal{Y}_i^{dom}, \\ 0, & \text{otherwise}. \end{cases} \tag{13}$$

- **Domain-Domain Edge** aims to construct direct relationships between domains at different hierarchical levels. For $\forall v_p^{dom}, v_q^{dom} \in \mathcal{G}$, the existence of the edge e_{pq} can be defined as:

$$e_{pq} = \begin{cases} 1, & \text{if } q \text{ is a direct subdomain of } p, \\ 0, & \text{otherwise}. \end{cases} \tag{14}$$

5.4. Feature enhancement

In order to integrate the news features and domain features in the network, the Feature Enhancement Network is used to aggregate and enhance node information. Specifically, a heterogeneous graph neural network is employed to learn node representations specific to each node type and facilitate message passing between each edge type.

The hidden state $\boldsymbol{h}_v^{(l+1)}$ of node v at the $(l+1)$th layer can be updated by aggregating messages from neighboring nodes. The process can be represented as:

$$\boldsymbol{h}_v^{(l+1)} = \sigma\left(\sum_{r\in\Psi} f_\theta^{(r)}\left(\boldsymbol{h}_v^{(l)}, \boldsymbol{h}_w^{(l)}\right)\right), w \in \mathcal{N}^{(r)}(v), \tag{15}$$

where $r \in \Psi$ denotes the type of edge, $f_\theta^{(r)}(\cdot)$ denotes the message passing function for each edge type, $\mathcal{N}^{(r)}(v)$ denotes the set of neighboring nodes to which node v is connected via edge r, and $\sigma(\cdot)$ denotes the activation function.

Since Graph Attention Network (GAT) (Brody et al., 2022; Veličković et al., 2018) can identify the neighbors that are more important to the current node, GAT was chosen as the basis in Eq. (15). For a particular relation r, GAT is used to obtain a representation of the node v:

$$\boldsymbol{h}_v'(H) = \bigoplus_{h=1}^{H} \sigma\left(\sum_{w\in\mathcal{N}(v)} \alpha_{vw}^h \boldsymbol{W}^h \boldsymbol{h}_w\right), \tag{16}$$

$$\alpha_{vw} = \mathrm{softmax}\left(\mathrm{LeakyReLU}\left(\boldsymbol{a}^{\mathrm{T}}\left[\boldsymbol{W}\boldsymbol{h}_v \oplus \boldsymbol{W}\boldsymbol{h}_w\right]\right)\right), \tag{17}$$

where H is the number of attention heads and $\boldsymbol{a}^{\mathrm{T}}$ is an attention mapping vector. Then the representation of v is subjected to an independent linear layer:

$$\boldsymbol{h}_v'' = \boldsymbol{W}\boldsymbol{h}_v + \boldsymbol{b}. \tag{18}$$

The final result of the message passing function is derived by aggregating the results of Eqs. (16) and (18):

$$f_\theta^{(r)}(\cdot) = \boldsymbol{h}_v'(H) + \boldsymbol{h}_v''. \tag{19}$$

Upon passing through L layers, the fused features of the news nodes and domain nodes can be obtained:

$$\boldsymbol{r}_{news}' = \boldsymbol{h}_{news}^{(L)}, \ \boldsymbol{r}_{dom}' = \boldsymbol{h}_{dom}^{(L)}. \tag{20}$$

5.5. Prediction with domain classifcation

Utilizing the enhanced node features as described in Eq. (20), it becomes feasible to predict the truth of the news, while domain nodes are additionally categorized to further enhance feature differentiation:

$$\hat{y} = \mathrm{softmax}\left(\boldsymbol{W}\boldsymbol{r}_{news}' + \boldsymbol{b}\right), \tag{21}$$

$$\hat{y}^{dom} = \mathrm{softmax}\left(\boldsymbol{W}\boldsymbol{r}_{dom}' + \boldsymbol{b}\right). \tag{22}$$

The objective is to minimize the cross-entropy loss associated with predicting the categories of both news and domain nodes. Finally, a composite loss is designed by combining these two losses ($\mathcal{L}_{news}$, $\mathcal{L}_{dom}$), yielding the overall loss function for the entire model:

$$\mathcal{L}_{news} = -y_i \log \hat{y}_i - \left(1 - y_i\right) \log\left(1 - \hat{y}_i\right), \tag{23}$$

$$\mathcal{L}_{dom} = -\sum_{d=1}^{|\Omega|} y_{id}^{dom} \log\left(\hat{y}_{id}^{dom}\right), \tag{24}$$

$$\mathcal{L} = \mathcal{L}_{news} + \mathcal{L}_{dom}, \tag{25}$$

where y_i and $\hat{y}_i$ are the ground-truth label and predicted label of news nodes respectively, y_{id}^{dom} and $\hat{y}_{id}^{dom}$ are the ground-truth label and predicted label of domain nodes respectively, and $|\Omega|$ represents the number of domain categories.

6. Experiments

6.1. Dataset

Based on DNT[2], this paper constructs a multi-domain multi-label enhanced dataset, which is more in line with the actual news classification situation and can fully describe the diversity of news. At the same time, in order to alleviate the dependence of the domain label annotation process on manual work, LLM is used to annotate when constructing the dataset. It is important to acknowledge that the labeling process is inherently prone to noise, particularly due to the illusions associated with LLM. Such noisy data can be detrimental to the performance of existing methods. A key aspect of our design is aimed at demonstrating that our proposed approach can mitigate the adverse effects caused by this noise. The construction process of the multi-domain multi-label enhanced dataset will be described in detail below.

Table 1
Data statistics of our dataset.

Domain	Politics	Economy	Medical	Edu.	Science	Sports	Env.	Military	Ent.	Society	All
#Real	1396	289	356	99	146	58	264	126	419	1707	2909
#Fake	2133	146	426	49	200	65	181	242	662	2076	3796
Total	3529	435	782	148	346	123	445	368	1081	3783	6705

6.1.1. Data collection

Fake and real news are all collected from PolitiFact and Snopes, spanning the period from ***January 2019*** to ***October 2023***. To preserve the effect of semantic learning, both fake and real news items are sourced from identical platforms. This approach diverges from certain existing datasets (Ayoub et al., 2021; Goldani et al., 2021; Samadi et al., 2021), where fake news is derived from fact-checking websites while real news is obtained from news outlets. Additionally, the selected samples present challenges in recognition and enhance the complexity of the task. Each news item includes: (1) **news content**, our dataset contains only text modality, which is the most important information for fake news detection; (2) **date**, we use the news release date if available otherwise we use the fact-check date; (3) **fack check result**, these results are given by human fact-checkers on these fact-checking websites; (4) **fact check URL**, because these fact-checking websites provide detailed evidence, providing verification links corresponding to news content can enhance credibility.

PolitiFact and Snopes employ disparate classifications for verifying news, so we harmonize them to suit the binary classification task. At the same time, we also consider the balance between true and false categories as much as possible. Specifically, ***true, mostly-true*** of PolitiFact and ***True, Mostly True*** of Snopes are unified under the label ***real***, while ***pants-fire*** of PolitiFact and ***False, Fake*** of Snopes are categorized as ***fake***. Samples that do not align with these categories are excluded from the dataset.

6.1.2. Data annotation

Since the news content has been manually verified by the above websites, we will mainly focus on the classification of domain labels. The annotation process follows these three fundamental rules:

1. Each sample may be assigned multiple domain labels.
2. When a sample is annotated with a subdomain, it must be recursively annotated with the parent domain of it.
3. If a sample pertains to a domain but does not fit any of its subdomains, it is labeled only with that domain.

Zero-shot prompting is employed to instruct GPT-3.5[3] (specifically, gpt-3.5-turbo-0125) in annotation according to the specified guidelines. In the process of news annotation, we initially assign the appropriate first-level labels to the news and subsequently annotate the next-level labels based on the results of the previous stage. The prompt we used in this process is as follows:

Prompt 1: Domain Annotation Prompt

Given content: `<CONTENT>`
Select the most appropriate topic categories in: `<DOMAIN>`
When answering, you only need to answer the topic category. If there are multiple matching ones, separate them with commas. Do not have other punctuation points. Do not answer beyond the options given. Think carefully before answering. If there is no matching topic category, answer 'unknown'.

where `<CONTENT>` represents the news to be annotated, and `<DOMAIN>` represents the list of labels at the current level. The output is filtered to remove recurring labels and those that fall outside the specified range. Subsequently, the parent domains of the involved domains are supplemented in accordance with the above rules. The statistics of our dataset are presented in Table 1. It should be noted that *All* in the table represents the total number of samples. However, because each sample can have multiple domain labels, this value does not equal the sum of samples across individual domains.

6.1.3. Data analysis

We conduct a preliminary analysis of each domain and its subdomains within the collected data to observe the distinctions and intrinsic connections among domains from a macro perspective. Fig. 4 illustrates the distribution of word clouds across different domains. Due to space limitations, only two major domains and their respective subdomains are presented here. It is particularly noteworthy that there are some words that appear in both two major domains with a higher frequency, such as *Donald Trump* and *COVID*. The analysis reveals that while the politics and medical domains exhibit considerable differences, there are both differences and similarities among the subdomains within the same domain. These observations highlight the inadequacy of the flat domain categorization employed in existing methodologies.

[3] https://openai.com/

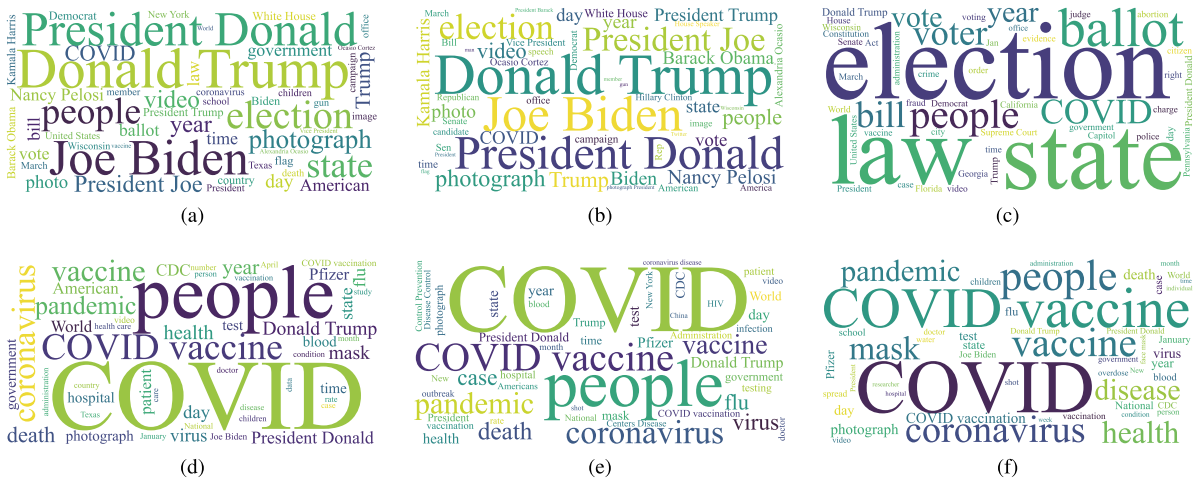


Fig. 4. Word clouds of different domains, where (a) is *Politics*, (b) is *Political Figures*, (c) is *Laws*, (d) is *Medical*, (e) is *Infectious Disease*, (f) is *Disease Prevention*.

Table 2
Performance of LLM annotation in different domains.

Domain	Politics	Economy	Medical	Edu.	Science	Sports	Env.	Military	Ent.	Society
Accuracy	0.8300	0.9800	0.9450	0.9900	0.9550	0.9700	0.9700	0.9900	0.9500	0.6950
Precision	0.8306	0.8973	0.9343	0.9349	0.7621	0.7374	0.7273	0.8974	0.8461	0.7333
Recall	0.8295	0.9535	0.8534	0.9349	0.7868	0.5808	0.9846	0.8974	0.9300	0.7111
F1	0.8297	0.9232	0.8875	0.9349	0.7738	0.6173	0.8047	0.8974	0.8816	0.6909

6.1.4. Evaluation of LLM annotation effects

To evaluate the effectiveness of zero-shot prompting in guiding LLM for multi-label annotation, we sample 200 news from the dataset for manual annotation. PolitiFact and Snopes each uniformly select 100 samples based on time. We report the Accuracy, macro-Precision, macro-Recall, and macro-F1 scores for the annotation results across each domain, as shown in Table 2.

According to Table 2, the F1 scores of most domains exceed 80%, demonstrating that LLMs exhibit good performance in domain annotation tasks. However, the *Science*, *Sports* and *Society* domains show notably lower performance. This phenomenon may be attributed to the limitations of these data sources (PolitiFact and Snopes), which result in insufficient sample coverage in those domains, consequently affecting the LLMs' processing capabilities.

In addition, to quantify the differences between LLM and human annotations, we introduce the Relative Coverage (RC) metric, adapted from Sørensen–Dice index. Specifically, for each sample i with its corresponding label sets L_i^{Human} and L_i^{LLM}, we defined bidirectional Relative Coverage as follows:

$$\begin{cases} RC_{\text{LLM}\to\text{Human}} = \dfrac{\left|L_i^{\text{LLM}} \cap L_i^{\text{Human}}\right|}{\left|L_i^{\text{Human}}\right|}, \\ RC_{\text{Human}\to\text{LLM}} = \dfrac{\left|L_i^{\text{LLM}} \cap L_i^{\text{Human}}\right|}{\left|L_i^{\text{LLM}}\right|}, \end{cases} \tag{26}$$

where, $RC_{X\to Y} \in [0,1]$, $RC_{\text{LLM}\to\text{Human}}$ measures the extent to which LLM annotations cover human annotations. In contrast, $RC_{\text{Human}\to\text{LLM}}$ assesses the extent to which human annotations cover LLM annotations. The closer these two values are, the higher the consistency between LLM and humans.

According to Eq. (26), we obtain $RC_{\text{LLM}\to\text{Human}} = \mathbf{0.7033}$ and $RC_{\text{Human}\to\text{LLM}} = \mathbf{0.6357}$. The higher value of $RC_{\text{LLM}\to\text{Human}}$ indicates that LLM tends to generate larger label sets with more labels compared to human annotations. Considering the results in Table 2 and Relative Coverage, although LLM annotation introduces some noise, the level remains within acceptable limits.

6.2. Baselines

We compare HFGD with three groups of state-of-the-art methods to evaluate the effectiveness, including two classic methods, three multi-domain methods, and four graph-based methods:

- TextCNN (Kim, 2014): This is a convolutional neural network based approach that extracts features from the token sequence of the input news content by using one-dimensional convolution.
- BiGRU (Ma et al., 2016): This is a widely used baseline in fake news detection, which uses a recurrent neural network to encode the tokens of news content.

- EANN (Wang et al., 2018): This method uses a multimodal feature extractor and an event discriminator to remove event-specific features and retain a common latent representation.
- MDFEND (Nan et al., 2021): This is a multi-domain fake news detection model based on MoE, which uses a gating mechanism to select a suitable expert network.
- M^3FEND (Zhu et al., 2023): This method uses domain information to aggregate representations of different views such as semantics, sentiment, and style.
- FCN-LP (Zhao et al., 2023): This method uses CLIP to extract multimodal features and uses the similarity of the two modalities to construct an homogeneous tweet graph.
- HeteroSGT (Zhang, Ma, et al., 2024): This method constructs a heterogeneous graph involving news articles, entities and topics, and employs a random walk and restart algorithm to extract subgraphs centered on each news for detection.
- MSynFD (Xiao et al., 2024): This method integrates a syntactical dependency graph, a multi-hop subgraph aggregation mechanism, and a keywords-based debiasing approach to capture complex syntactic relations in fake news.
- HFGD*: This method is a variant of HFGD, which excludes the domain node classification loss $\mathcal{L}_{dom}$ due to the paucity of studies directly utilizing news relations for graph construction.

6.3. Implementation details

The proposed method is implemented using PyTorch and PyG, with all parameters initialized stochastically. The dataset is divided into training, validation, and test sets with the ratio of 6:2:2. A mini-batch size of 64 is employed for all methods, and no fine-tuning is performed. The selected Sentence-BERT is all-mpnet-base-v2, and the maximum sequence length of sentences is uniformly set to 170. For a fair comparison, the same Sentence-BERT model is utilized as the semantic extraction model for all methods, replacing the BERT and RoBERTa instance. The news semantic similarity threshold τ is set to 0.6. Domain node features in heterogeneous networks are computed only from the training set, while all domain nodes are preserved in all sets. In the Gaussian kernel pooling process, we use $K = 512$ kernels with $\sigma_k = 5$, where the means μ_k are uniformly sampled from [0, 1]. A two-layer heterogeneous GAT network with a hidden dimension of 640 is employed in feature enhancement. The learning rate for Adam optimizer is 1e-4.

All TextCNN used in baselines have 5 kernels with steps in {1, 2, 3, 5, 10} and 64 channels. The hidden dimension of all MLPs in baseline methods is set to 384, with the ReLU activation function employed. Since only text modality is available, for a fair comparison, the image modality in the multimodal method is omitted. For BiGRU, we use a single-layer implementation with a hidden dimension of 300. For EANN, since the domain in our dataset is multi-label, the loss function of the event discriminator is modified to a multi-label cross-entropy loss to adapt to our scenario. For M^3FEND, since we only use the semantic information of news, this information is fed into the feature extraction network for all three views simultaneously, without removing the sentiment and style components. For FCN-LP, Maximum Mean Discrepancy (MMD) loss is removed as it does not match the scenario we set up. EANN, MDFEND, and M^3FEND use domain labels but assign single label to each news, so we adapt them by replicating each sample based on its multiple domain labels. For HeteroSGT, we extract entities and topics in the same way as described in the original paper, and the number of topics extracted by the Latent Dirichlet Allocation (LDA) model is set to 44 for a fair comparison. For MSynFD, we obtain syntactical dependency graph and keywords of news following the same methodology in their paper. The learning rates for TextCNN, BiGRU, EANN, MDFEND, M^3FEND, FCN-LP, HeteroSGT, and MSynFD are set to 1e-4, 1e-3, 1e-4, 7e-5, 1e-4, 3e-4, 5e-5, and 1e-5, respectively.

Except for the mentioned parameters, we reused the code and model parameters provided in the original papers. All models are trained using an NVIDIA GeForce RTX 4090 GPU on our proposed dataset. We select all models with the best performance on the validation set for testing. Each model is trained 10 times, and we report the average and standard deviation of the results.

6.4. Evaluation metrics

To evaluate the performance of all models, we employ four widely used metrics in fake news detection: Accuracy, macro-Precision, macro-Recall, and macro-F1. Accuracy provides a general view of performance but may not be sufficient for imbalanced data. Macro-Precision, macro-Recall, and macro-F1 calculate the average of the respective scores across all classes, weighing each class equally regardless of its size. These metrics ensure equal contribution from minority classes, preventing majority class dominance in the result.

6.5. Overall performance

In this section, we conduct baseline experiments to demonstrate the effectiveness of our proposed method. Table 3 presents the overall performance of all methods, while Table 4 provides the metrics of multi-domain methods across different domains, respectively. We only present the results for the 10 major domains due to space constraints. The bold and underlined text indicates the best and second-best results, respectively.

(1) According to Table 3, the well designed multi-domain and graph-based methods generally achieve superior results compared to classical methods. BiGRU and TextCNN perform similarly, and their overall difference is not significant. This highlight the inadequacy of traditional methods in multi-domain scenarios. EANN's performance is lower than classical methods by 0.97% in F1 score, while MDFEND and M^3FEND show significant improvements of 1.73% and 2.53%, respectively. This improvement indicates the advantage of augmenting news features with domain information over learning domain-invariant features, consistent

Table 3
Performance comparison in general.

Method		Accuracy	Precision	Recall	F1
Classic	TextCNN	0.6941 ± 0.0089	0.6932 ± 0.0091	0.6859 ± 0.0177	0.6844 ± 0.0158
	BiGRU	0.7013 ± 0.0119	0.6966 ± 0.0122	0.6949 ± 0.0132	0.6949 ± 0.0127
Multi	EANN	0.6986 ± 0.0092	0.6957 ± 0.0073	0.6838 ± 0.0134	0.6852 ± 0.0137
	MDFEND	0.7184 ± 0.0045	0.7139 ± 0.0046	0.7118 ± 0.0072	0.7122 ± 0.0062
	M^3FEND	0.7221 ± 0.0069	0.7224 ± 0.0060	0.7250 ± 0.0062	0.7202 ± 0.0062
Graph	FCN-LP	0.7228 ± 0.0058	0.7180 ± 0.0058	0.7161 ± 0.0071	0.7168 ± 0.0067
	HeteroSGT	0.7083 ± 0.0072	0.7054 ± 0.0071	0.6939 ± 0.0084	0.6960 ± 0.0085
	MSynFD	0.7275 ± 0.0108	0.7234 ± 0.0111	0.7178 ± 0.0118	0.7195 ± 0.0117
	HFGD*	0.7321 ± 0.0030	0.7277 ± 0.0031	0.7237 ± 0.0032	0.7251 ± 0.0031
	HFGD	$\mathbf{0.7336 \pm 0.0061}$	$\mathbf{0.7299 \pm 0.0054}$	$\mathbf{0.7302 \pm 0.0053}$	$\mathbf{0.7293 \pm 0.0055}$

Table 4
Performance comparison in different domains (44 domain, multi-label).

Method	Politics				Economy				Medical				Education				Science			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
EANN	0.7171	0.7050	0.6846	0.6890	0.7440	0.7019	0.7124	0.7049	0.7209	0.7415	0.7105	0.7072	0.6000	0.5517	0.5550	0.5518	0.6519	0.6543	0.6447	0.6416
MDFEND	0.7530	0.7413	0.7352	0.7372	0.7560	0.7244	0.7438	0.7254	0.7439	0.7533	0.7370	0.7369	0.6600	0.6258	0.6200	0.6176	0.6494	0.6502	0.6423	0.6411
M^3FEND	0.7428	0.7343	0.7396	0.7339	**0.8070**	**0.7821**	**0.7640**	**0.7678**	**0.7719**	**0.7749**	**0.7722**	**0.7707**	0.6833	0.6376	0.6150	0.6169	**0.6922**	**0.6920**	**0.6922**	**0.6913**
HFGD	**0.7648**	**0.7533**	**0.7512**	**0.7520**	0.7950	0.7569	0.7602	0.7578	0.7288	0.7336	0.7229	0.7230	**0.7467**	**0.7380**	**0.6650**	**0.6750**	0.6597	0.6637	0.6595	0.6568

Method	Sports				Environment				Military				Entertainment				Society			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
EANN	0.4692	0.4983	0.4986	0.4610	0.5894	0.5926	0.5924	0.5881	0.7667	0.7447	0.7309	0.7361	0.6519	0.6654	0.6357	0.6282	0.7138	0.7094	0.7042	0.7054
MDFEND	0.4846	0.4890	0.4958	0.4669	0.6511	0.6488	0.6441	0.6440	0.7974	0.7787	0.7841	0.7796	0.6689	0.6678	0.6617	0.6615	0.7233	0.7203	0.7224	0.7201
M^3FEND	**0.5692**	**0.5319**	**0.5361**	**0.5238**	0.6681	0.6853	0.6516	0.6448	0.7833	0.7663	0.7803	0.7680	**0.6929**	0.6924	**0.6924**	**0.6912**	0.7184	0.7224	0.7252	0.7176
HFGD	0.4923	0.5144	0.5153	0.4773	**0.6851**	**0.7161**	**0.6657**	**0.6566**	**0.8346**	**0.8161**	**0.8273**	**0.8206**	0.6910	**0.6929**	0.6839	0.6830	**0.7313**	**0.7289**	**0.7316**	**0.7288**

Table 5
Performance comparison of different components.

	news rel.	domain rel.	domain feat.	domain loss	Accuracy	Precision	Recall	F1
(i)	✓	✓	✓		0.7321	0.7277	0.7237	0.7251
(ii)	✓	✓		✓	0.7013	0.6970	0.6963	0.6959
(iii)	✓		✓	✓	0.7078	0.7064	0.7088	0.7053
(iv)		✓	✓	✓	0.7100	0.7060	0.7062	0.7053
(v)	✓	✓	✓	✓	**0.7336**	**0.7299**	**0.7302**	**0.7293**

with previous research (Tang et al., 2020; Zhu et al., 2019). Graph-based methods outperform most non-graph methods, which supports the potential of multi-relational heterogeneous networks in fake news detection. Notably, the performance of HeteroSGT demonstrates inferior results compared to other graph-based methods. This underperformance may be attributed to its LDA-based topic extraction method, which excessively relies on word frequency distributions and overemphasizes high-frequency words.

(2) According to Table 4, HFGD significantly outperforms other multi-domain methods in most domains, enhancing both overall and domain-specific performance. In the *Sports* domain, all methods exhibit relatively low performance, which can be attributed to both its limited sample size and poor annotation quality referring to Table 2. In contrast, despite relatively inaccurate annotation in the *Society* domain, HFGD achieves an F1 score of 72.88% that primarily due to its large sample size. Notably, in the *Education* and *Military* domains, our methods surpass the second-best by 5.74% and 4.1% in F1 score respectively, despite their limited sample sizes. This improvement can be attributed to two factors: the semantic proximity of these two domains to *Society* and *Politics* which allows model to benefit from more training data in these related domains, and both *Education* and *Military* domains demonstrate high annotation accuracy as shown in Table 2.

These results demonstrate the effectiveness of our approach in leveraging domain information to explicitly model news-domain multi-relationships.

6.6. Ablation study

In this section, we perform ablation studies to demonstrate the effectiveness of the various components of HFGD: (i) remove the domain classification loss, (ii) replace domain features with a set of randomly initialized learnable vectors, (iii) remove the edges between domains, (iv) remove the edges between news items, (v) complete method. The results are shown in Table 5.

(1) First, we eliminate the domain node classification loss $\mathcal{L}_{dom}$ from the overall loss function. It is important to highlight that our domain classification is performed on domain nodes rather than directly categorizing the news articles themselves. This methodology

Table 6
Domain label effectiveness comparison.

Method		Accuracy	Precision	Recall	F1
10/S	EANN	0.6941 ± 0.0176	0.6894 ± 0.0178	0.6813 ± 0.0208	0.6827 ± 0.0208
	MDFEND	0.7199 ± 0.0061	0.7152 ± 0.0062	0.7129 ± 0.0063	0.7136 ± 0.0062
	M^3FEND	0.7208 ± 0.0131	**0.7208 ± 0.0114**	**0.7238 ± 0.0116**	**0.7191 ± 0.0125**
	HFGD	**0.7216 ± 0.0042**	0.7188 ± 0.0040	0.7204 ± 0.0058	0.7182 ± 0.0047
10/M	EANN	0.6856 ± 0.0187	0.6821 ± 0.0188	0.6793 ± 0.0209	0.6784 ± 0.0205
	MDFEND	0.7247 ± 0.0072	0.7199 ± 0.0072	0.7191 ± 0.0085	0.7193 ± 0.0081
	M^3FEND	0.7177 ± 0.0091	0.7183 ± 0.0109	0.7218 ± 0.0111	0.7164 ± 0.0097
	HFGD	**0.7254 ± 0.0054**	**0.7236 ± 0.0045**	**0.7253 ± 0.0064**	**0.7224 ± 0.0055**
44/M	EANN	0.6986 ± 0.0092	0.6957 ± 0.0073	0.6838 ± 0.0134	0.6852 ± 0.0137
	MDFEND	0.7184 ± 0.0045	0.7139 ± 0.0046	0.7118 ± 0.0072	0.7122 ± 0.0062
	M^3FEND	0.7221 ± 0.0069	0.7224 ± 0.0060	0.7250 ± 0.0062	0.7202 ± 0.0062
	HFGD†	0.7078 ± 0.0110	0.7064 ± 0.0100	0.7088 ± 0.0093	0.7053 ± 0.0101
	HFGD	**0.7336 ± 0.0061**	**0.7299 ± 0.0054**	**0.7302 ± 0.0053**	**0.7293 ± 0.0055**

distinguishes itself from other domain classification techniques, such as EANN, which focus on learning domain-invariant features. The results of (i) and (v) suggest that the auxiliary task of categorizing domain information helps distinguish between domains and improves detection capability.

(2) Next, we replace our meticulously designed domain features with a collection of randomly initialized learnable vectors. Comparing the results of (ii) and (v), we observe a significant drop in all metrics, e.g., the F1 score decreasing by 3.34%. This demonstrates the effectiveness of our well-designed domain features, indicating that they enhance the model's ability to understand news content.

(3) Finally, we removed the relationship edges connecting domains at different levels, as well as the relationship edges between different news items within the constructed heterogeneous network. Comparing the results in (iii), (iv), and (v), we observe a 2.39%–2.58% decrease in all metrics. Our approach leverages these varied relations to improve model performance, with the results demonstrating that constructing multi-relationships effectively enhances news representations.

After comprehensively comparing all the above variants, it becomes evident that the component exerting the most significant influence on performance is the well-designed domain features. This is followed by the diverse news-domain relationships, with the domain classification loss having the least impact. These conclusions demonstrate the importance of domain information and explicit modeling of multiple relationships in multi-domain fake news detection.

6.7. Domain label effectiveness analysis

In this section, we conduct experiments to assess the influence of domain labels, thereby demonstrating the effectiveness of the proposed DNT^2. To achieve this, we utilize three different domain label settings while ensuring consistency in other aspects, such as data partitioning. The specific configurations of the domain labels are outlined as follows:

- **10 domain, single-label (10/S)**: 10 major domains are used, with each sample assigned a unique domain label from our full dataset. Given that our dataset is structured for multi-label, we preferentially retain the domain with the smaller size to ensure diversity during selection.
- **10 domain, multi-label (10/M)**: 10 major domains are used, with each sample assigned multiple domain labels. This is achieved by removing all subdomain labels from our full dataset.
- **44 domain, multi-label (44/M)**: This is our full dataset which includes domain relationships. It is important to note that this setting does not merely increase the number of domains. To simulate a directly increase in the number of domains, we introduce a variant of our model, HFGD†, which removes the edges between domain nodes in this setup.

We report the overall accuracy, precision, recall, and macro F1 score of the multi-domain methods under these different settings, as shown in Table 6. Additionally, we report the metrics for the 10 major domains of these methods under the 10/S and 10/M settings, as shown in Tables 7 and 8, respectively, while the results for the 44/M setting are provided in the previous Table 4. From these results, we can draw the following conclusions:

(1) After introducing multi-label and domain segmentation, HFGD demonstrates substantial performance improvements both overall and across most domains. This is attributed to the increased precision and completeness of domain information which benefits relationship modeling. In contrast, other methods experience limited improvements. This phenomenon is likely due to the fact that these methods are not specifically designed for multi-label scenarios. In other words, converting multi-label data into single-label format often leads to redundancy due to the copy operations, so the performance of these methods is not as expected. Furthermore, the comparison between HFGD and HFGD† suggests that simply increasing the number of domains does not significantly enhance model performance.

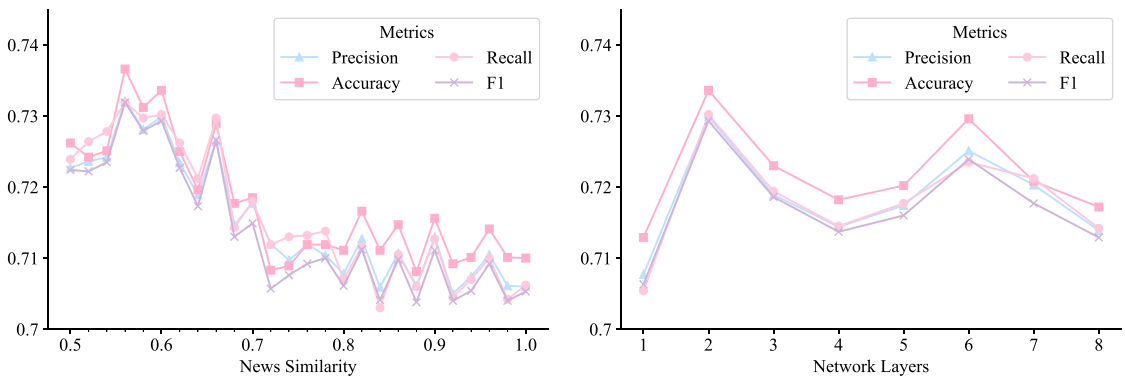
(2) In cases where domain information is annotated by automatic tools, previous methods cannot significantly enhance the detection accuracy, primarily due to their heavy reliance on precise domain annotations. Conversely, HFGD maintains its

Table 7
Performance comparison in different domains (10 domain, single-label).

Method	Politics				Economy				Medical				Education				Science			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
EANN	0.7301	0.6928	0.6805	0.6851	0.7333	0.6936	0.6994	0.6923	0.7248	**0.7533**	0.7103	0.7064	0.6267	0.5740	0.5700	0.5693	0.6416	0.6422	0.6367	0.6353
MDFEND	**0.7403**	0.7068	0.7002	0.7022	0.7417	0.7059	0.7289	0.7120	0.7333	0.7338	0.7302	0.7302	0.6267	0.5666	0.5600	0.5600	0.6753	**0.6787**	0.6684	0.6668
M^3FEND	0.7303	0.7048	0.7164	0.7059	**0.8115**	0.7854	0.7544	0.7651	**0.7368**	0.7379	**0.7356**	**0.7344**	0.6800	0.6338	0.6050	0.6088	**0.6779**	0.6783	**0.6766**	**0.6754**
HFGD	0.7299	**0.7102**	**0.7293**	**0.7115**	0.8083	**0.7859**	**0.7678**	**0.7706**	0.7231	0.7385	0.7110	0.7097	**0.7433**	**0.7341**	**0.6700**	**0.6783**	0.6377	0.6552	0.6242	0.6127
Method	Sports				Environment				Military				Entertainment				Society			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
EANN	0.4769	0.4978	0.4972	0.4659	0.5807	0.5863	0.5855	0.5794	0.7737	0.7436	0.7437	0.7431	0.6625	0.6686	0.6490	0.6454	0.6766	0.6756	0.6775	0.6723
MDFEND	**0.5462**	**0.5476**	**0.5542**	**0.5244**	0.6506	0.6542	0.6409	0.6394	**0.8395**	**0.8175**	**0.8315**	**0.8225**	0.6979	0.6991	0.6900	0.6898	**0.7109**	**0.7116**	**0.7137**	**0.7076**
M^3FEND	0.5346	0.5171	0.5181	0.5039	0.6747	0.6932	0.6585	0.6527	0.7895	0.7731	0.7952	0.7746	0.7125	0.7111	0.7108	0.7104	0.6897	0.7095	0.7073	0.6893
HFGD	0.4231	0.4617	0.4653	0.4066	**0.6892**	**0.6944**	**0.6787**	**0.6778**	0.8382	0.8162	0.8244	0.8187	**0.7240**	**0.7227**	**0.7198**	**0.7204**	0.6909	0.7053	0.7049	0.6899

Table 8
Performance comparison in different domains (10 domain, multi-label).

Method	Politics				Economy				Medical				Education				Science			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
EANN	0.6954	0.6811	0.6765	0.6766	0.7160	0.6655	0.6562	0.6566	0.7209	0.7311	0.7133	0.7119	0.6200	0.5538	0.5500	0.5478	0.6571	0.6612	0.6550	0.6528
MDFEND	**0.7497**	**0.7374**	0.7333	**0.7348**	0.7560	0.7218	0.7495	0.7285	**0.7669**	**0.7680**	**0.7654**	**0.7653**	0.6133	0.5691	0.5650	0.5648	**0.6883**	**0.6922**	**0.6829**	**0.6813**
M^3FEND	0.7386	0.7311	**0.7400**	0.7318	**0.8180**	**0.7900**	**0.7643**	**0.7734**	0.7583	0.7584	0.7582	0.7574	**0.7167**	**0.6920**	**0.6525**	**0.6598**	0.6740	0.6737	0.6727	0.6722
HFGD	0.7424	0.7316	0.7359	0.7323	0.7810	0.7439	0.7540	0.7457	0.7331	0.7413	0.7258	0.7257	0.7100	0.6754	0.6325	0.6393	0.6584	0.6581	0.6554	0.6543
Method	Sports				Environment				Military				Entertainment				Society			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
EANN	0.5154	0.5096	0.5111	0.4900	0.5894	0.5918	0.5873	0.5804	0.7667	0.7450	0.7484	0.7447	0.6698	0.6735	0.6600	0.6573	0.6873	0.6861	0.6866	0.6835
MDFEND	**0.5615**	**0.5698**	**0.5792**	**0.5440**	0.6553	0.6574	0.6444	0.6428	0.8103	0.7898	0.7991	0.7934	0.6708	0.6686	0.6649	0.6647	0.7241	0.7203	0.7226	0.7209
M^3FEND	0.5154	0.4947	0.4938	0.4833	0.6723	0.6961	0.6543	0.6463	0.7833	0.7743	0.7960	0.7725	0.6887	0.6872	0.6859	0.6857	0.7134	0.7195	0.7219	0.7130
HFGD	0.4346	0.4922	0.4944	0.4279	**0.6947**	**0.7199**	**0.6792**	**0.6739**	**0.8269**	**0.8083**	**0.8232**	**0.8132**	**0.6981**	**0.6973**	**0.6964**	**0.6958**	**0.7255**	**0.7285**	**0.7304**	**0.7239**



(a) Parameter τ (b) Parameter L

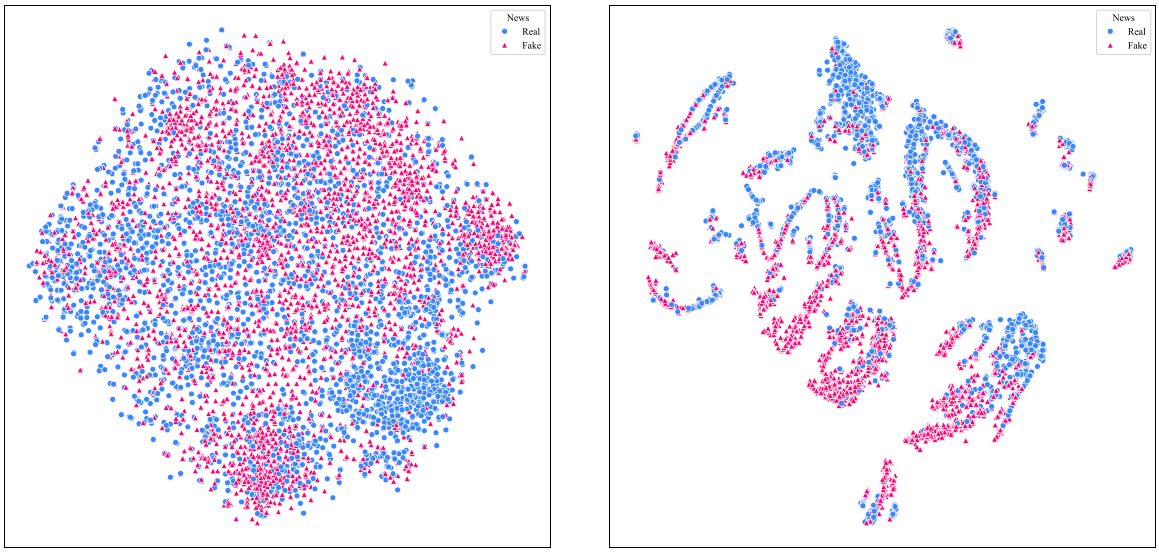
Fig. 5. Parameters sensitivity.

effectiveness even with automatically generated labels. This robustness arises from explicit multi-relationship modeling, which allows the model to use news information in related domains for better accuracy and reduces constraints on label quality.

6.8. Parameters sensitivity

In this section, we investigate the sensitivity of similarity threshold τ and the number of feature enhancement network layers L. The results are presented in Fig. 5, where Fig. 5(a) is the result of the news semantic similarity threshold, and Fig. 5(b) is the result of the number of feature enhancement network layers.

(1) The similarity threshold τ plays a crucial role in constructing the graph by determining the connections between news nodes. A higher threshold ensures more reliable connections but leads to a sparser and potentially less informative graph, while a lower



(a) Original semantic features (b) Learned features

Fig. 6. Feature visualization.

threshold captures more potential relationships but may introduce noise and increase complexity. To balance graph connectivity and information quality, we evaluate τ from 0.5 to 1.0 (with a step size of 0.02). As shown in Fig. 5(a), the model achieves better detection performance when τ is around 0.6. However, the performance undergoes a decline of approximately 2% when τ exceeds 0.7, which is attributed to the overly sparse graph structure caused by the high similarity threshold. Based on these observations, we set the final hyperparameter τ to 0.6.

(2) The number of feature enhancement network layers L determines the range of neighbor information that a node can aggregate. We evaluate L from 1 to 8. As shown in Fig. 5(b), the model exhibits a saddle-like performance pattern, achieving the best results when $L = 2$ and showing a secondary peak at $L = 6$. When $L = 2$, most news nodes can achieve satisfactory results by aggregating information through indirect relationships. The performance rebound at $L = 6$ suggests that a small portion of nodes benefit from longer aggregation paths. However, given that $L = 2$ has provided good performance in most cases, and considering the computational overhead associated with deeper layers, we set the final hyperparameter L to 2.

6.9. Visualization

In Fig. 6, we employ t-SNE to reduce the dimension of news node features to 2D for visualization. Fig. 6(a) displays the visualization of the original semantic features of the news text, and Fig. 6(b) illustrates the learned features after the model integrates domain information for news nodes. It can be found that the original feature distribution reveals a mixture of real and fake samples, with some exhibiting a slight clustering tendency. In contrast, the learned features demonstrate more pronounced boundaries. Furthermore, in Fig. 6(b), it can be observed that the representation of news nodes, augmented by domain features, establishes distinct clusters at the macro level. Some clusters are positioned at considerable distances from one another, while others are situated in close proximity. At the micro level, real and fake samples within clusters are clearly distinguishable. We propose that these clusters reflect the relationships among different domains: the closer the distance between domains, the stronger their relationship is. This indicates that incorporating domain information effectively groups news with similar meanings. However, some domains exhibit blurred boundaries between real and fake samples, highlighting a limitation of classification based solely on semantic information.

6.10. Time complexity analysis

Since the method proposed in this paper introduces a complex graph structure, we conduct a time complexity analysis in this section. All time tests are conducted on the aforementioned GPU device and Intel Xeon Silver 4210 @ 2.20 GHz CPU. First, we compare the training time of all baseline methods, as shown in Fig. 7. The proposed HFGD segregates the total time into heterogeneous graph construction and neural network training stages.

The experimental results demonstrate that methods based on complex structures require longer training time, particularly those introducing heterogeneous graph structures (HeterSGT, MSynFD, and HFGD). In order to further analyze the computational cost of HFGD during heterogeneous graph construction, we evaluate its time complexity from a theoretical perspective.

The primary computational burden in HFGD lies in the calculation of domain structural features, which requires iterative analysis of each domain graph. Considering a dataset comprising $|\Omega|$ domains, where each domain graph consists of N nodes and M edges. According to Algorithm 1, the time complexity analysis can be broken down into the following steps:

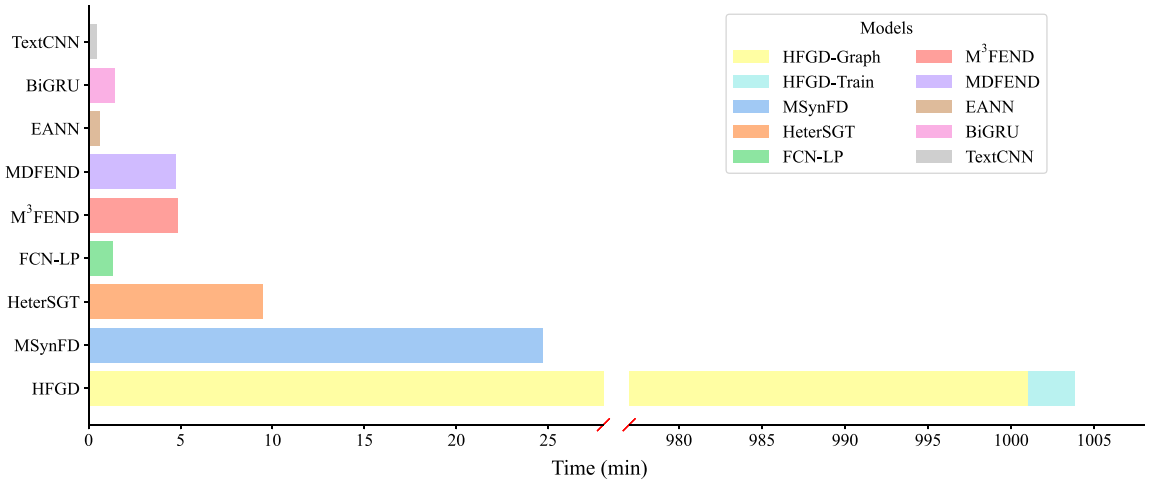


Fig. 7. Training time of all compared methods.

- For betweenness centrality calculation (Algorithm 1, Line 9): The Brandes algorithm is employed to compute the betweenness centrality of all nodes in the weighted graph. First, Dijkstra's algorithm is used to compute the weighted shortest path for each node, with a single-instance time complexity of $O(M + N \log N)$. Then the shortest path dependency value is computed from the current node to all other nodes, necessitating access to all edges and nodes, yielding a single-instance time complexity of $O(N + M)$. Therefore, the total time complexity for computing all N nodes is:

$$N \times (O(M + N \log N) + O(N + M)) = O\left(NM + N^2 \log N\right). \tag{27}$$

- For Shannon diversity index calculation (Algorithm 1, Line 10–20): The process of sorting nodes based on betweenness centrality exhibits a time complexity of $O(N \log N)$. Node deletion operations require $O(N)$ time, while connected component computation needs $O(N + M)$ time. Organizing nodes by connected component size demands $O(N \log N)$ time, and the Shannon diversity index computation for remaining nodes requires $O(N)$ time. Therefore, the aggregate time complexity for computing all N nodes is:

$$O(N \log N) + N \times (O(N) + O(N + M) + O(N \log N) + O(N)) = O\left(NM + N^2 \log N\right). \tag{28}$$

- The graph structure analyzed in this study is sparse ($M \approx N$), consequently, the overall time complexity for processing a single domain graph is:

$$O\left(NM + N^2 \log N\right) + O\left(NM + N^2 \log N\right) = O\left(N^2 \log N\right) + O\left(N^2 \log N\right) = O\left(N^2 \log N\right). \tag{29}$$

  Finally, the total time complexity for processing all $|\Omega|$ domain graphs is:

$$O\left(|\Omega| N^2 \log N\right) = O\left(N^2 \log N\right). \tag{30}$$

Since this process involves full-sample calculations, it incurs significant computational overhead with large-scale datasets. However, this computational cost is nevertheless essential for capturing complex inter-feature relationships, thus providing a richer information representation for the model. Notably, although the construction process of heterogeneous graph in HFGD is time-intensive, it constitutes a one-time preprocessing overhead. The resultant graph structure can be preserved and utilized in subsequent training and inference phases without additional computational costs. Furthermore, HFGD currently employs a serial processing approach for heterogeneous graph construction on the CPU, with significant dependence on processor performance. Future optimizations could incorporate parallel computing and other computational strategies to improve construction efficiency.

7. Discussion and implications

In our research on multi-domain fake news detection, we propose the Domain-Specific Narrow-Coverage Tree-Based Taxonomy (DNT^2) and the Hierarchical Fine-Grained Domain (HFGD) Fake News Detection Method, which aim to address the limitations of previous studies by offering a novel perspective.

Firstly, we observe that almost all existing methods require datasets to have precise domain labels, which limits their real-world applications. This limitation arises because the annotation process of domain labels inevitably introduces potential biases, and manual domain annotation is costly. To address this issue, DNT^2 further refines domains into more detailed categories, which enables a more nuanced understanding of news content within specific domains and mitigates domain bias. Concurrently, the utilization

of LLMs for automatic annotation diminishes the reliance on manual labor and facilitates the development of scalable fake news detection systems. The obtained multi-domain multi-label enhanced dataset captures the diversity of cross-domain news content and offers resources for training robust detection models. Additionally, most of these methods classify news into a single domain, which is generally applicable but has evident limitations. In contrast, the HFGD method employs a heterogeneous network to explicitly model multi-domain relationships for each sample, trying to cover various possible situations. This approach enables domains with fewer samples to assimilate information from larger ones, thereby improving the model's robustness to domain bias.

Our work has several important practical implications. First, the proposed DNT^2 taxonomy and HFGD method provide a more comprehensive and accurate framework for fake news detection across multiple domains, which can enhance existing fact-checking platforms and social media monitoring systems. Second, our approach of using LLMs for automatic domain annotation provides a solution for reducing manual annotation costs, making it valuable for organizations with limited human resources. Third, our enhanced multi-domain dataset can serve as a valuable benchmark for future research in cross-domain fake news detection and domain adaptation studies. Finally, our findings highlight the importance of considering domain relationships in fake news detection, which could inform the design of future content moderation systems.

It is important to acknowledge the limitations of this study. While our dataset and methodology yield promising results, they are confined to specific domains and news sources. From a computational perspective, our approach requires considerable computational resources when processing large-scale datasets, and future optimizations through parallel processing and other computational strategies could further enhance its scalability and efficiency. Several important directions warrant further investigation in the future work. A primary focus will be incorporating knowledge evolution techniques to track changes in domain entities and relationships over time. This enhancement will enable the DNT^2 system to dynamically adapt to emerging domains. Another crucial direction is improving real-time processing capabilities, particularly for social media environments where timely detection is essential. This will involve optimizing the HFGD method to efficiently handle streaming data. Beyond these technical improvements, we aim to explore the generalizability of HFGD across other domains and languages. Finally, mitigating potential biases introduced by LLMs during the annotation process remains an important research direction to ensure the robustness and reliability of our approach.

In summary, this study introduces a novel and practical approach for multi-domain fake news detection. However, further research is necessary to enhance the robustness of this framework in more complex scenarios.

8. Conclusion

This study conducts a comprehensive analysis of two principal challenges in multi-domain fake news detection: (1) the high costs associated with domain annotation and the limited resistance to label noise, and (2) the issues of sparsity and bias in domain labels. To address these challenges, we introduce the Domain-Specific Narrow-Coverage Tree-Based Taxonomy (DNT^2), which mitigates domain bias through the refinement of domain categories, achieving more precise domain classification and better cross-domain relationship modeling compared to traditional approaches. Additionally, we utilize zero-shot prompting to guide LLMs for annotation that significantly reduces the reliance on manual labor while maintaining annotation quality, and create a multi-domain multi-label enhanced dataset that effectively captures the diversity of news. Furthermore, we propose the Hierarchical Fine-Grained Domain (HFGD) Fake News Detection Method, which improves the robustness to noise and ensures substantial distinctions between domains by explicitly modeling multi-relationships within a heterogeneous network. Experimental results indicate that HFGD surpasses existing techniques in both accuracy and annotation efficiency, particularly in mitigating domain bias in underrepresented domains such as education and military. These improvements make the proposed method valuable for practical fact-checking systems and content moderation platforms, especially when dealing with cross-domain or data-scarce domain content. Future work will explore the potential of LLMs further to enhance the depth and breadth of fake news detection.

CRediT authorship contribution statement

Yue Wang: Writing – original draft, Software, Methodology, Data curation. **Shizhong Yuan:** Software, Methodology. **Weimin Li:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yifan Feng:** Writing – review & editing, Methodology, Data curation. **Xiao Yu:** Writing – review & editing, Visualization, Validation. **Fangfang Liu:** Writing – review & editing. **Can Wang:** Writing – review & editing. **Quanke Pan:** Writing – review & editing.

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Data availability

Data will be made available on request.

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